

Fine-Tuning PaLI-Gemma and MedGemma for Unified Multimodal Medical Imaging Tasks

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Abstract—Vision-language foundation models (VLMs) can solve many different medical imaging tasks using the same model, by training them on both images and text and then tuning them. In this work, we fine-tune and benchmark two VLMs—PaLI-Gemma-2 and MedGemma-4B-IT—using a single unified medical imaging dataset. This dataset combines 19 public datasets across 8 medical imaging modalities, and includes 50,996 images with 58,112 question-answer pairs. We evaluate both models on a wide range of tasks, including classification, regression, counting, detection, and instance-level detection. PaLI-Gemma is fine-tuned using a JAX/Flax training pipeline with attention-only training and masked token-level cross-entropy loss. MedGemma is fine-tuned using LoRA (Low-Rank Adaptation) for parameter-efficient training, implemented with Hugging Face Transformers and TRL, using a chat-style instruction format. Our results show that classification is the strongest task for both models (PaLI-Gemma achieves about 60–62% accuracy, while MedGemma reaches 65.73%). However, regression and counting remain difficult, mainly because the models struggle with precise numerical reasoning. For localization tasks, PaLI-Gemma often generates incorrectly formatted JSON outputs, which reduces evaluation reliability. In contrast, MedGemma produces more stable structured outputs and achieves strong performance in predicting coordinates (detection R^2 up to 0.93 and instance detection $R^2 = 0.7247$). Overall, the study suggests that medical-domain instruction tuning improves the reliability and controllability of VLM outputs, especially for structured medical imaging tasks.

Index Terms—Vision-language models, medical imaging, multimodal learning, fine-tuning, LoRA, structured prediction.

I. INTRODUCTION

Medical imaging is very important in healthcare. Doctors use it for screening diseases, diagnosing patients, and monitoring health over time. Today’s hospitals generate many different types of medical images, such as X-rays, ultrasound scans, endoscopy images, microscope images, retinal images, skin images, and mammograms.

However, building machine learning models that work well across all these image types is still difficult. This is mainly because images from different modalities look very different, hospitals follow different scanning protocols and patient populations, so the data distribution changes, and medical imaging tasks are diverse ranging from simple classification (for example, detecting if something is abnormal) to localization tasks (like drawing bounding boxes around lesions) and numerical prediction tasks (like counting cells or estimating a continuous measurement).

Traditionally, researchers train separate deep learning models for each task, such as CNN classifiers, U-Net segmentation models, or object detection models like Faster R-CNN and YOLO. Although these approaches often work well, they are not scalable. They usually require a large amount of labeled data for every new task, and they are limited because they cannot easily combine image understanding with medical language reasoning and knowledge.

Vision-language foundation models (VLMs) provide a promising alternative to traditional medical imaging models. Because they are trained on large amounts of both images and text, a single VLM can be adapted to many different tasks using prompts and small amounts of fine-tuning. This motivates us to study whether a general-purpose VLM (PaLI-Gemma) and a medical-domain instruction-tuned VLM (MedGemma) can handle different medical imaging tasks in one unified framework.

Contributions:

- We fine-tune and evaluate PaLI-Gemma-2 and MedGemma-4B-IT on a unified medical imaging benchmark that includes classification, regression, counting, detection, and instance detection tasks.
- We compare two practical fine-tuning strategies: attention-only fine-tuning for PaLI-Gemma and parameter-efficient LoRA instruction tuning for MedGemma, under limited computing resources.
- We present a detailed analysis of where the models fail, especially in terms of numerical reasoning, structured output correctness, and unstable localization predictions.

II. RELATED WORK

A. Vision-language models

Vision-language models (VLMs) jointly process images and text to support multimodal understanding and generation. Most VLMs consist of (i) an image encoder (e.g., Vision Transformer) that extracts visual features and (ii) a Transformer-based language decoder that generates text responses. Image and text features are typically fused using cross-attention or related fusion mechanisms. Large-scale multimodal pretraining enables VLMs to generalize to many downstream tasks via prompting and fine-tuning.

B. Medical multimodal learning

Applying general-purpose VLMs to medical imaging is challenging due to domain shift between natural and clinical images and the risk of hallucinations in generated text. To address this, medical VLMs are trained using clinical images together with biomedical/clinical text, improving domain alignment and reliability. Instruction tuning further improves controllability by teaching models to follow task-specific instructions and produce structured outputs (e.g., class labels, coordinates, or JSON).

C. Parameter-efficient fine-tuning

Fine-tuning large VLMs is computationally expensive because they contain billions of parameters. Parameter-efficient fine-tuning (PEFT) methods reduce this cost by freezing the base model and training only a small set of additional parameters. LoRA is a widely used PEFT method that learns low-rank updates in attention layers, significantly reducing GPU memory usage and training time. This makes LoRA practical for fine-tuning large medical VLMs under limited compute resources.

III. DATASET AND TASK FORMULATION

We use the dataset from the Unified Multimodal Medical Imaging Challenge containing 19 datasets across 8 imaging modalities, comprising 50,996 images and 58,112 question-answer pairs. Tasks include classification, regression, counting, detection, and instance detection.

A. Task types

Let I denote an image and q the natural-language question prompt. The model predicts a textual output $\hat{y}$.

Classification: $\hat{y} \in \mathcal{C}$ where $\mathcal{C}$ is label set.

Regression: $\hat{y} \in \mathbb{R}$ (scalar represented as text).

Counting: $\hat{y} \in \mathbb{Z}_{\geq 0}$.

Detection: $\hat{y}$ is a structured tuple $(x_{min}, y_{min}, x_{max}, y_{max})$.

Instance detection: $\hat{y}$ is a structured mapping $\{class_id : [[x, y, w, h], \dots]]\}$.

TABLE I
DATASET OVERVIEW.

Property	Value
Modalities	8
Datasets	19
Images	50,996
QA pairs	58,112
Task types	Cls/Reg/Count/Det/InstDet

IV. METHODOLOGY

A. PaLI-Gemma-2 architecture

PaLI-Gemma is a pretrained multimodal transformer. It integrates a vision encoder that maps images into patch embeddings, and a language decoder which generates output tokens. Multimodal fusion occurs through cross-attention or injected visual tokens, allowing the decoder to condition on image features.

B. PaLI-Gemma attention-only fine-tuning

We adopt a partial fine-tuning strategy to reduce compute. Only attention sublayers of the LLM decoder are updated:

- query projection W_q ,
- key/value projection W_k, W_v ,
- attention output projection W_o .

All other parameters remain frozen. Training is performed using JAX/Flax (big_vision pipeline), which supports multi-device sharding and efficient input loading.

1) *Masked token-level objective:* We use a prefix-suffix tokenization approach:

- Prefix: input instruction prompt (loss masked out)
- Suffix: expected model response (loss enabled)

2) *Masked next-token cross entropy Loss:* Given a batch of size B , each training sample consists of an image I_b and a tokenized text sequence $y_b = \{y_{b,1}, y_{b,2}, \dots, y_{b,T}\}$ of length T . The model is trained autoregressively: at each time step t , it predicts the next token $y_{b,t+1}$ conditioned on the image I_b and the previous tokens $y_{b,\leq t}$. Let $p_\theta(\cdot)$ denote the probability distribution produced by the model (after softmax) with trainable parameters θ . To exclude padding tokens and prompt/prefix tokens from optimization, we apply a binary loss mask $m_{b,t} \in \{0, 1\}$, where $m_{b,t} = 1$ indicates tokens that belong to the answer and should contribute to the loss. The per-sample loss is computed as the average negative log-likelihood over valid (unmasked) tokens, and the final training loss is obtained by averaging across the batch:

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^B \frac{-\sum_{t=1}^{T-1} m_{b,t} \log p_\theta(y_{b,t+1} | y_{b,\leq t}, I_b, q_b)}{\sum_{t=1}^{T-1} m_{b,t}} \quad (1)$$

Where,

B is the batch size,

T is the token sequence length,

I_b is the input image for sample b ,

q_b is the question/prompt text,

$y_{b,t}$ denotes the token at position t , and

$m_{b,t}$ is the loss mask (0 for prefix, 1 for answer tokens).

This masking ensures that training optimizes response generation rather than reproducing the question text.

C. MedGemma-4B-IT architecture

MedGemma is a transformer-based medical VLM pretrained and instruction tuned using medical images and clinical language. The instruction-tuned variant (IT) is optimized for responding to user prompts with structured and clinically relevant outputs.

D. LoRA fine-tuning for MedGemma

We perform LoRA tuning due to large base model size. LoRA injects trainable low-rank matrices A, B into selected linear layers:

$$W' = W + \Delta W, \quad \Delta W = BA, \quad \text{rank}(\Delta W) = r \quad (2)$$

where $r \ll d$. This reduces number of trainable parameters while maintaining adaptation capability.

We use LoRA rank $r = 16$ and $\alpha = 16$, targeting linear modules. The base model weights remain frozen.

1) *Chat-style formatting*: Each training example is converted to chat messages:

- User: <image> + instruction/question
- Assistant: reference response/answer

Padding tokens and image tokens are masked with -100 in labels so they do not contribute to the loss.

V. EXPERIMENTAL SETUP

A. Data preprocessing

All images are converted to RGB, resized to 224×224 , and normalized. Grayscale images are repeated across 3 channels; alpha channels are removed. Invalid image paths are removed. Data is shuffled and split into train/validation/test (80/10/10).

B. Training configuration

PaLI-Gemma uses cosine LR schedule with 10% warmup and SGD optimizer; LR in the range 5×10^{-5} to 5×10^{-4} with batch size 8 or 16 depending on experiment. MedGemma uses AdamW with LR 2×10^{-4} , LoRA adapters, bfloat16 precision, gradient checkpointing, and warmup.

TABLE II
TRAINING CONFIGURATIONS USED IN EXPERIMENTS.

Setting	PaLI-Gemma-2	MedGemma-4B-IT
Framework	JAX/Flax	HF + TRL
Fine-tuning	Attention-only	LoRA
Batch size	8 or 16	4 (with accumulation)
Learning rate	$5e-5$ – $5e-4$	$2e-4$
Scheduler	Cosine + warmup	Linear + warmup
Precision	fp32/fp16	bfloat16
Epochs/steps	Multi runs	~ 5621 steps

C. Evaluation metrics

Classification uses exact-match accuracy. Regression and counting are evaluated using:

$$MAE = \frac{1}{N} \sum |y_i - \hat{y}_i|, \quad (3)$$

$$RMSE = \sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2} \quad (4)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (5)$$

For detection, we compute MAE and R^2 over coordinates and also report number of malformed structured outputs.

D. Implementation and Compute

All experiments were conducted on the IIT Delhi HPC infrastructure using NVIDIA GPUs (e.g., A100 80GB / V100 32GB depending on availability). Training was performed with mixed precision where supported (bfloat16 for MedGemma) to reduce memory footprint and accelerate throughput.

VI. RESULTS AND DISCUSSION

A. Classification

PaLI-Gemma models 1–6 achieve consistent classification accuracy (60–62%), while model 7 underperforms (48–50%), suggesting sensitivity to hyperparameter selection. MedGemma achieves higher accuracy (65.73%) reflecting benefits of medical-domain instruction tuning.

TABLE III
CLASSIFICATION ACCURACY (EXACT MATCH).

Model	Val Acc. (%)	Test Acc. (%)
PaLI-Gemma-2 (best models)	~ 62.0	~ 62.0
MedGemma-4B-IT (LoRA)	67.14	65.73

B. Regression

PaLI-Gemma regression exhibits negative/near-zero R^2 despite stable MAE/RMSE. This implies predictions are close to constant mean estimates and do not explain variance. This can occur due to: (i) numeric information being poorly represented in language token space, (ii) insufficient task-specific supervision, and (iii) heterogeneous label scales across modalities.

PaLI-Gemma Model 2 achieves the highest test R^2 of 0.0004, which is negligible. MedGemma improves MAE and RMSE but still has low $R^2 = 0.035$, confirming numerical regression remains challenging.

TABLE IV
PALI-GEMMA REGRESSION SUMMARY (FROM BEST-PERFORMING RUNS).

PaLI Model	Test MAE	Test RMSE	R^2
Model 2 (best R^2)	13.40	18.26	0.0004
Model 5 (low MAE)	13.18	18.33	-0.01

TABLE V
MEDGEMMA REGRESSION RESULTS.

Model	MAE	RMSE	R^2
MedGemma-4B-IT (LoRA)	9.84	15.78	0.035

C. Counting

PaLI-Gemma counting similarly yields mostly negative R^2 . Model 5 is best with test $R^2 = 0.17$. This suggests partial trend learning but poor exact discrete estimation. MedGemma improves $R^2 = 0.365$, though exact count accuracy remains low.

TABLE VI
PALI-GEMMA COUNTING SUMMARY.

Model	Test MAE	Test RMSE	R^2
Model 5 (best)	58.5	94.26	0.17

TABLE VII
MEDGEMMA COUNTING RESULTS.

Model	MAE	RMSE	R^2
MedGemma-4B-IT (LoRA)	47.6	78.35	0.365

D. Detection and instance detection

Detection tasks require structured coordinate output. PaLI-Gemma often fails to output valid JSON structures, producing 49 malformed outputs in training and 36 in testing. Such formatting failures make evaluation unreliable and indicate weak instruction following in structured generation.

MedGemma produces better structured outputs and achieves coordinate-level R^2 up to 0.93. Instance detection also improves strongly with $R^2 = 0.7247$, indicating most coordinate variance is explained.

TABLE VIII
MALFORMED OUTPUTS.

Model	Train	Test
PaLI-Gemma-2	49	36
MedGemma-4B-IT	very low	very low

VII. ERROR ANALYSIS

To better understand the strengths and weaknesses of both models, we analyze the main failure patterns observed across tasks. We identify three dominant issues:

(1) Weak numerical grounding. Regression and counting require accurate understanding of numbers (e.g., “12.5”, “3 cells”, “left ventricle diameter”). However, VLMs generate numbers as text tokens, and the training objective mainly encourages fluent language generation rather than precise numerical correctness. As a result, both PaLI-Gemma and MedGemma frequently produce rounded or near-constant estimates, leading to low or negative R^2 even when MAE/RMSE appears reasonable.

(2) Structured output errors in localization tasks. Detection and instance detection require the model to output coordinates in a strict format (e.g., JSON). We observe that PaLI-Gemma often generates malformed structures, such as missing brackets, incorrect key names, or invalid numeric fields. These small formatting mistakes break parsing and reduce evaluation reliability. In contrast, MedGemma shows stronger instruction-following behavior and produces more consistent structured outputs, which improves coordinate-level evaluation scores.

(3) Task imbalance and optimization bias. The unified dataset contains heterogeneous tasks, but their learning difficulty is not uniform. Classification tasks provide more stable gradients and clear supervision, while regression/counting/localization require more fine-grained reasoning and strict output constraints. In multi-task training, this can bias optimization toward easier tasks such as classification, leading to weaker performance on structured and numeric tasks.

Overall, these observations suggest that current VLM fine-tuning pipelines are more effective for recognition-style tasks (classification) than for precise numerical or structured prediction without additional constraints or specialized training strategies.

VIII. LIMITATIONS AND FUTURE WORK

Although the experiments provide useful insights into unified multimodal fine-tuning, the study has several limitations.

Limitations. First, our evaluation focuses on exact-match and numeric metrics, which may not fully capture semantic correctness. For example, a model may generate a medically meaningful answer with slightly different wording and still be marked incorrect. Second, structured tasks (detection and instance detection) are highly sensitive to output formatting. Even minor JSON errors result in failed parsing, which underestimates the true visual understanding capability of the model. Third, both models were trained under practical compute constraints (attention-only fine-tuning for PaLI-Gemma and LoRA for MedGemma), which may limit the achievable performance compared to full fine-tuning.

Future work. Several directions could improve reliability and performance on difficult tasks: (i) introducing constrained decoding or schema-guided generation to guarantee valid JSON outputs for localization tasks; (ii) adding output validation and automatic correction for malformed structured predictions; (iii) improving numerical reasoning using discretized regression bins, special numeric tokens, or auxiliary numeric loss terms; (iv) designing task-conditioned prompts and training strategies to reduce multi-task imbalance; (v) exploring stronger adaptation strategies such as partial unfreezing of MLP layers or using modality-specific adapters.

These improvements could significantly strengthen VLM reliability for practical medical imaging deployment, especially for structured prediction tasks.

IX. CONCLUSION

This work presented a unified fine-tuning and benchmarking study of PaLI-Gemma-2 and MedGemma-4B-IT on a multimodal medical imaging benchmark spanning 19 datasets across 8 imaging modalities. We evaluated both models on five task families: classification, regression, counting, detection, and instance detection.

Across experiments, classification emerged as the most reliable capability for both models, with MedGemma outperforming PaLI-Gemma due to medical-domain instruction tuning. In contrast, regression and counting remained challenging for both models, highlighting the limited numerical grounding of generative vision-language architectures. For structured localization tasks, we observed a major difference in output stability: PaLI-Gemma frequently produced malformed structured outputs, while MedGemma generated more valid coordinate predictions and achieved strong coordinate-level quality.

Overall, our findings suggest that medical instruction tuning and parameter-efficient adaptation improve controllability

and structured generation, making instruction-tuned medical VLMs more suitable for unified medical imaging settings. At the same time, the results highlight the need for improved numerical supervision and constrained decoding methods to enable reliable regression and structured prediction.

CODE AND MODEL AVAILABILITY

The code, training scripts, and notebooks used in this study are publicly available at: <https://github.com/Shobhit-2510/Model-Training-for-Multimodal-Medical-Imaging/tree/main>. The fine-tuned model checkpoints are available on Hugging Face: https://huggingface.co/ssshobhit1106/Project_2025/tree/main.

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REFERENCES

- [1] E. Hu, Y. Shen, P. Wallis, et al., “LoRA: Low-Rank Adaptation of Large Language Models,” *arXiv:2106.09685*, 2021.
- [2] Google AI for Developers, “Fine-tune PaLI-Gemma,” 2024. [Online]. Available: <https://ai.google.dev/gemma/docs/paligemmma/fine-tuning-paligemmma>. Accessed: July, 2025.
- [3] Google Health, “MedGemma: Fine-tune with Hugging Face (Notebook),” 2024. [Online]. Available: https://colab.research.google.com/.../fine_tune_with_hugging_face.ipynb. Accessed: Sep. 2025.
- [4] Hugging Face, “Model Card: google/paligemmma-3b-mix-224,” 2024. [Online]. Available: <https://huggingface.co/google/paligemmma-3b-mix-224>. Accessed: July, 2025.
- [5] Hugging Face, “Model Card: google/medgemma-4b-it,” 2024. [Online]. Available: <https://huggingface.co/google/medgemma-4b-it>. Accessed: Oct. 2025.
- [6] L. Beyer et al., “PaliGemma: A versatile 3B VLM for transfer,” *arXiv preprint arXiv:2407.07726*, 2024.
- [7] A. Steiner et al., “PaliGemma 2: A Family of Versatile VLMs for Transfer,” *arXiv preprint arXiv:2412.03555*, 2024.
- [8] A. Sellergren et al., “MedGemma Technical Report,” *arXiv preprint arXiv:2507.05201*, 2025.
- [9] CodaBench, “MICCAI FLARE 2025 Task 5: Multimodal Model for Medical ... (Competition 7151),” 2025. [Online]. Available: <https://www.codabench.org/competitions/7151/>. Accessed: July, 2025.